hyper2kvm

Enterprise Storage: LVM & LUKS Auto-Detection

Automatic detection and handling of LVM, LUKS encryption, and complex storage stacks. Supermin appliance boots for full device visibility. Zero configuration needed.

Verified: RHEL 8.8 + CentOS 9 + Ubuntu 22.04 LUKS

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The Problem

Enterprise VMs use complex storage stacks that break during migration

LVM Volumes

- Root filesystem on /dev/mapper/vg-root
- VG activation requires device-mapper inside appliance
- UUID regeneration for cloned VMs
- dracut needs --add lvm dm for boot

LUKS Encryption

- Root on LUKS+LVM (/dev/sda3 → crypt → /dev/vg/root)
- cryptsetup needs full device visibility
- initramfs must NOT be rebuilt (breaks UUID refs)
- TPM-sealed keys don't transfer between hypervisors

Standard Approach Limitations

- Host-based qemu-nbd + mount works for simple disks
- LVM via container isolation — works but complex
- LUKS via host cryptsetup — device paths differ
- No isolated appliance = limited device visibility

hyper2kvm Solution

- Auto-detect LVM/LUKS before opening disk
- Boot supermin appliance for full isolation
- cryptsetup + LVM activation inside appliance
- Zero user configuration needed

How It Works

Automatic backend selection based on disk content

```
Disk arrives
     |
Quick probe: qemu-nbd --read-only + blkid scan
     |
LVM/LUKS found? --yes--> Supermin appliance boots
                          cryptsetup_open()  --> LUKS unlock
                          lvm_scan()         --> LVM activation
                          mount()            --> mount decrypted root
                          command(dracut)    --> rebuild initramfs (non-LUKS only)

                --no---> VMCraft (fast path, no appliance)
                         qemu-nbd --> host mount --> chroot dracut
```

Key Design Decision

For LUKS disks: **skip initramfs rebuild**. The guest's existing initramfs already has cryptsetup support. Rebuilding inside the appliance breaks LUKS UUID references because device mapper paths differ from what the guest expects.

LUKS Migration Pipeline

Ubuntu 22.04 with LUKS2 + LVM + UEFI — fully automated

```
$ sudo h2kvmctl --cmd local \
    --vmdk ~/vmware/ubuntu-encrypted.vmdk \
    --luks-enable --luks-passphrase "redhat123" \
    --flatten --regen-initramfs \
    --emit-domain-xml --virsh-define --vm-name ubuntu-luks

Quick probe: detected LUKS on disk
Auto-switched to supermin appliance backend
LUKS: opened /dev/sda3 -> /dev/mapper/hyper2kvm-crypt1
LVM after LUKS open: ['/dev/ubuntu-vg/ubuntu-lv']
Detected guest: Ubuntu 22.04.5 LTS
LUKS detected (crypttab exists) -- preserving existing boot config
Skipping initramfs rebuild (LUKS disk)
Virtio driver configs written to /etc/dracut.conf.d/
Offline fixes complete
EFI System Partition detected -- auto-enabling UEFI
Domain 'ubuntu-luks' defined + started

# VM boots --> "Please unlock disk dm_crypt-0:" --> enter passphrase --> login
```

LVM Migration Pipeline

RHEL 8.8 with LVM thin provisioning — fully automated

```
$ sudo h2kvmctl --config test-confs/test-config.yaml

Quick probe: detected LVM on disk
Auto-switched to supermin appliance backend
LVM detected: VGs=['rhel'], LVs=['/dev/rhel/root', '/dev/rhel/swap']
Running dracut -f --kver 4.18.0-432.el8.x86_64 \
    --add-drivers virtio_blk virtio_scsi virtio_net nvme ahci sd_mod xts \
    --add lvm dm
Offline fixes complete
Domain 'rhel8.8-migrated' defined + started
# VM boots to GDM login screen
```

For non-LUKS LVM disks, initramfs IS rebuilt with virtio + LVM modules. dracut runs inside the supermin appliance with full device-mapper visibility.

Verified Configurations

All tested end-to-end: VMDK → flatten → fix → compress → libvirt → boot

Guest OS	Storage	Detection	initramfs	Boot	Result
RHEL 8.8 Beta	LVM thin (VG: rhel)	Auto (LVM)	dracut rebuilt	BIOS	**GDM login**
CentOS Stream 9	LVM (VG: cs)	Auto (LVM)	dracut rebuilt	BIOS	**Console login**
Ubuntu 22.04.5	LUKS2 + LVM	Auto (LUKS)	Preserved	UEFI	**Passphrase prompt**

Storage Operations

LUKS Unlock

- cryptsetup_open(device, key, name)
- clevis_luks_unlock (TPM)
- Passphrase, keyfile, or env var
- LVM rescan after unlock

LVM Activation

- pvscan --cache --activate ay
- vgscan + vg_activate_all
- UUID regeneration for clones
- Thin provisioning support

Guest Fix Operations

- Mount decrypted root filesystem
- Fix fstab, network, grub
- dracut with virtio+LVM modules
- SELinux autorelabel

Architecture: Supermin Appliance

Isolated VM boots for full device visibility — same proven technique used in enterprise migration tools

Supermin Appliance

- Boots host kernel + minimal rootfs (~30MB)
- Full udevd for device discovery
- LVM: pvscan --cache --activate ay
- cryptsetup for LUKS unlock
- Communicates via virtio-serial

Auto-Detection Flow

- Quick probe: qemu-nbd + blkid
- Scans partitions for LVM2_member / crypto_LUKS
- Auto-switches backend before opening disk
- Falls back to VMCraft for simple disks
- Zero user configuration needed

LUKS: What We Preserve

- /etc/crypttab — preserved as-is
- initramfs — NOT rebuilt (UUID integrity)
- GRUB cryptdevice= — preserved
- Virtio configs written for post-boot rebuild

TPM-Sealed LUKS

- VMware vTPM ≠ KVM vTPM (different seeds)
- Pre-migration: add passphrase keyslot
- clevis luks unbind to remove TPM binding
- Post-migration: re-bind to KVM vTPM
- Script: v2v-luks-tpm-prepare.sh

CLI Reference

LUKS migration commands

```
# Passphrase-based LUKS
sudo h2kvmctl --cmd local --vmdk disk.vmdk \
  --luks-enable --luks-passphrase "password" \
  --flatten --regen-initramfs \
  --emit-domain-xml --virsh-define -o /output

# Passphrase from environment
export HYPER2KVM_LUKS_PASSPHRASE="password"
sudo h2kvmctl --cmd local --vmdk disk.vmdk --luks-enable ...

# Keyfile
sudo h2kvmctl --cmd local --vmdk disk.vmdk \
  --luks-enable --luks-keyfile /path/to/keyfile ...

# YAML config
cat << EOF > migration.yaml
cmd: local
vmdk: /path/to/encrypted.vmdk
luks_enable: true
luks_passphrase: "password"
flatten: true
regen_initramfs: true
emit_domain_xml: true
virsh_define: true
vm_name: my-luks-vm
EOF
sudo h2kvmctl --config migration.yaml
```

Summary

Enterprise Storage Support: Complete

LVM, LUKS, LUKS+LVM, UEFI, thin provisioning — all verified end-to-end

What's New

- Auto-detect LVM/LUKS via quick probe
- Auto-switch to supermin appliance
- LUKS unlock + LVM-inside-LUKS
- Preserve initramfs for LUKS disks
- TPM preparation script included

Tested

- RHEL 8.8 LVM thin → GDM
- CentOS 9 LVM → console
- Ubuntu 22.04 LUKS+LVM+UEFI
- Split VMDK (8 extents)
- 1271 unit tests passing

Production Ready

- Supermin appliance (proven 15+ years)
- Zero configuration for users
- Automatic fallback for simple disks
- RPM + DEB packages updated
- Full documentation + guides